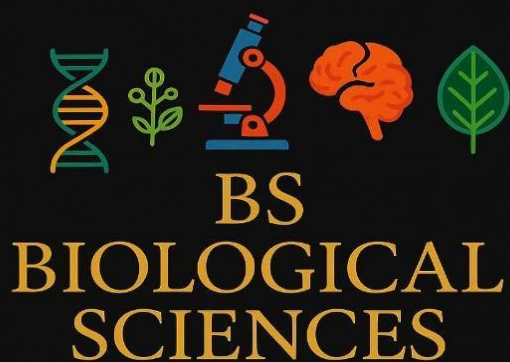




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BIF401P- Bioinformatics-I (Practical)

BIF401P FINAL-TERM OBJECTIVES

PREPARATION FILE

By

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BIF401P FINAL-TERM OBJECTIVES PREPARATION FILE:

Practical No. 01

1. Genetic information in most living organisms is stored in:

- A) Protein
- B) DNA
- C) Lipid
- D) Carbohydrate
- **Answer: B**

2. In RNA, thymine (T) is replaced by:

- A) Adenine
- B) Guanine
- C) Uracil
- D) Cytosine
- **Answer: C**

3. DNA contains how many nitrogenous bases?

- A) 2
- B) 3
- C) 4
- D) 5
- **Answer: C**

4. Proteins are made up of:

- A) Fatty acids
- B) Nucleotides

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- C) Amino acids
- D) Sugars
- **Answer: C**

5. NCBI was established in:

- A) 1985
- B) 1988
- C) 1992
- D) 1995
- **Answer: B**

6. NCBI was established by the:

- A) WHO
- B) NIH
- C) UNESCO
- D) FAO
- **Answer: B**

7. Which database stores DNA and RNA sequences?

- A) UniProt
- B) PDB
- C) GenBank
- D) InterPro
- **Answer: C**

8. Which database stores protein information?

- A) GenBank
- B) UniProt
- C) BLAST
- D) PDB

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- **Answer: B**

9. GenBank is maintained by:

- A) NIH
- B) EMBL
- C) DDBJ
- D) SIB

- **Answer: A**

10. Which of the following is NOT an NCBI database/tool?

- A) BLAST
- B) OMIM
- C) dbSNP
- D) ClustalW

- **Answer: D**

11. GenBank data can be downloaded in:

- A) FASTA
- B) XML
- C) TXT
- D) All of these

- **Answer: D**

Practical No. 02

1. UniProt is mainly used for:

- A) DNA sequences
- B) Protein sequences
- C) RNA editing

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- D) Gene cloning
 - **Answer: B**
2. UniProt stands for:
- A) Universal Protein Resource
 - B) Universal Protein Repository
 - C) Unique Protein Resource
 - D) Universal Peptide Resource
 - **Answer: A**
3. UniProt is produced by:
- A) NCBI
 - B) UniProt Consortium
 - C) EMBL
 - D) DDBJ
 - **Answer: B**
4. Which is the manually reviewed section of UniProtKB?
- A) TrEMBL
 - B) Swiss-Prot
 - C) UniParc
 - D) UniMES
 - **Answer: B**
5. Which database contains automatically annotated protein sequences?
- A) Swiss-Prot
 - B) TrEMBL
 - C) UniRef
 - D) PDB
 - **Answer: B**

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6. UniRef90 contains sequences that are:
- A) 50% identical
 - B) 70% identical
 - C) 90% or more identical
 - D) 100% different
 - **Answer: C**
7. UniRef100 contains sequences that are:
- A) 100% identical
 - B) 90% identical
 - C) 80% identical
 - D) 50% identical
 - **Answer: A**
8. Which UniProt database removes duplicate protein sequences?
- A) UniMES
 - B) UniParc
 - C) Swiss-Prot
 - D) TrEMBL
 - **Answer: B**
9. UniMES contains protein sequences from:
- A) Human genome
 - B) Environmental samples
 - C) Viruses only
 - D) Plants only
 - **Answer: B**
10. Which organization is NOT part of the UniProt Consortium?
- A) EBI

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- B) PIR
- C) SIB
- D) NIH
- **Answer: D**

11. The largest public protein database is:

- A) GenBank
 - B) PDB
 - C) UniProt
 - D) BLAST
 - **Answer: C**
-

Practical No. 03

1. Sequence alignment compares:

- A) DNA only
- B) RNA only
- C) Proteins only
- D) DNA, RNA, or proteins
- **Answer: D**

2. Global alignment aligns sequences:

- A) Partially
- B) End to end
- C) Randomly
- D) Locally only
- **Answer: B**

3. Needleman-Wunsch is used for:

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- A) Local alignment
- B) Global alignment
- C) Phylogeny
- D) Protein modeling

• **Answer: B**

4. Smith-Waterman algorithm performs:

- A) Global alignment
- B) Local alignment
- C) MSA
- D) Protein prediction

• **Answer: B**

5. Dynamic programming is used to:

- A) Build proteins
- B) Solve alignment problems efficiently
- C) Sequence DNA
- D) Predict mutations

• **Answer: B**

6. Sequence alignment helps identify:

- A) Unknown genes
- B) Similar proteins
- C) Evolutionary relationships
- D) All of these

• **Answer: D**

7. Which alignment finds similar regions only?

- A) Global
- B) Local

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- C) Circular
- D) Progressive

• **Answer: B**

8. Sequence alignment can predict:

- A) Gene function
- B) Protein function
- C) Evolution
- D) All of these

• **Answer: D**

9. Homology studies commonly use:

- A) PCR
- B) Sequence alignment
- C) ELISA
- D) Microscopy

• **Answer: B**

10. Physical and genetic maps are built using:

- A) Alignment
- B) Centrifugation
- C) Electrophoresis
- D) Staining

• **Answer: A**

11. Dynamic programming works by:

- A) Ignoring smaller problems
- B) Breaking problems into smaller parts
- C) Sequencing DNA
- D) Copying proteins

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- **Answer: B**

Practical No. 04

1. MSA stands for:

- A) Molecular Sequence Analysis
- B) Multiple Sequence Alignment
- C) Mutation Sequence Analysis
- D) Multiple Structure Alignment

- **Answer: B**

2. MSA aligns:

- A) Two sequences
- B) Three or more sequences
- C) One sequence
- D) Protein structures only

- **Answer: B**

3. MSA is an extension of:

- A) PCR
- B) Pairwise alignment
- C) Sequencing
- D) Gene mapping

- **Answer: B**

4. Which tool is commonly used for MSA?

- A) ClustalW
- B) MUSCLE
- C) MAFFT

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- D) All of these

• **Answer:** D

5. MSA helps identify:

- A) Conserved regions
- B) Random mutations only
- C) DNA extraction methods
- D) Restriction enzymes

• **Answer:** A

6. MSA is useful for constructing:

- A) PCR primers
- B) Phylogenetic trees
- C) Agarose gels
- D) Plasmids

• **Answer:** B

7. ClustalW follows which alignment strategy?

- A) Random
- B) Progressive
- C) Local only
- D) Manual

• **Answer:** B

8. The first step in ClustalW is:

- A) Guide tree construction
- B) Pairwise alignment
- C) Distance calculation
- D) Final alignment

• **Answer:** B

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9. ClustalW builds a guide tree using:

- A) Maximum likelihood
- B) Neighbor-joining
- C) UPGMA only
- D) Bayesian analysis
- **Answer: B**

10. A limitation of ClustalW is:

- A) Supports only local alignment
- B) Cannot align proteins
- C) May fail with long insertions
- D) Cannot build guide trees
- **Answer: C**

11. Consensus sequences are obtained using:

- A) PCR
- B) BLAST
- C) MSA
- D) PDB
- **Answer: C**

Practical No. 05

1. BLAST stands for:

- A) Basic Local Alignment Search Tool
- B) Biological Local Analysis System Tool
- C) Basic Link Analysis Software Tool
- D) Bioinformatics Local Alignment Tool

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- **Answer: A**
2. BLAST is mainly used to:
- A) Predict protein structure
 - B) Compare DNA or protein sequences
 - C) Build phylogenetic trees
 - D) Extract DNA
- **Answer: B**
3. The sequence entered into BLAST is called the:
- A) Subject
 - B) Template
 - C) Query
 - D) Target
- **Answer: C**
4. BLAST helps identify:
- A) Similar genes
 - B) Similar proteins
 - C) Unknown sequences
 - D) All of these
- **Answer: D**
5. Which scoring matrices are used in BLAST?
- A) PAM and BLOSUM
 - B) FASTA and XML
 - C) DNA and RNA
 - D) GMQE and TM-score
- **Answer: A**
6. Which BLAST program compares nucleotide sequences?

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- A) BLASTP
- B) BLASTN
- C) BLASTX
- D) TBLASTN
- **Answer: B**

7. Which BLAST program compares protein sequences?

- A) BLASTN
- B) BLASTP
- C) BLASTX
- D) TBLASTX
- **Answer: B**

8. Which is NOT a BLAST output section?

- A) Alignments
- B) Taxonomy
- C) Graphic Summary
- D) ProtParam
- **Answer: D**

9. BLAST is widely used because it is:

- A) Slow
- B) Fast
- C) Manual
- D) Offline only
- **Answer: B**

10. BLASTX compares:

- A) Protein vs Protein
- B) DNA translated into protein vs protein database

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- C) DNA vs DNA
- D) RNA vs RNA
- **Answer: B**

11. BLAST is mainly used in:

- A) Bioinformatics
 - B) Histology
 - C) Anatomy
 - D) Ecology
 - **Answer: A**
-

Practical No. 06

1. A phylogenetic tree mainly represents:

- A) Protein structure
- B) Evolutionary relationships
- C) DNA extraction
- D) PCR products
- **Answer: B**

2. The organisms or sequences shown in a phylogenetic tree are called:

- A) Nodes
- B) Taxa
- C) Branches
- D) Leaves
- **Answer: B**

3. A single organism or sequence in a tree is called:

- A) Taxon
- B) Taxa

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- C) Branch

- D) Root

- **Answer: A**

4. Branch length represents:

- A) DNA size

- B) Evolutionary change

- C) Protein weight

- D) Number of genes

- **Answer: B**

5. Two closely related taxa are called:

- A) Root groups

- B) Sister groups

- C) Families

- D) Domains

- **Answer: B**

6. A clade consists of:

- A) Unrelated species

- B) A common ancestor and its descendants

- C) Different kingdoms

- D) Random sequences

- **Answer: B**

7. The root node represents:

- A) Latest species

- B) Common ancestor

- C) Mutation site

- D) Protein sequence

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- **Answer: B**

8. Bootstrap values indicate:

- A) Sequence length
- B) Confidence level of branching
- C) Protein stability
- D) Gene expression

- **Answer: B**

9. Evolutionary distance is represented by:

- A) Node color
- B) AA + BB branch lengths
- C) Taxon names
- D) Sequence ID

- **Answer: B**

10. Internal nodes usually represent:

- A) Modern species
- B) Common ancestors
- C) Mutations
- D) Chromosomes

- **Answer: B**

11. Phylogenetic trees are useful for studying:

- A) Evolution
- B) Classification
- C) Relationships among species
- D) All of these

- **Answer: D**

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Practical No. 07

1. In a phylogenetic tree, branches represent:
 - A) Evolutionary lineages
 - B) DNA fragments
 - C) Proteins
 - D) Chromosomes
 - **Answer: A**
2. Nodes in a phylogenetic tree indicate:
 - A) DNA mutations only
 - B) Common ancestors
 - C) Protein domains
 - D) RNA sequences
 - **Answer: B**
3. Closely related organisms are usually connected by:
 - A) Longer branches
 - B) Shorter evolutionary distance
 - C) Random nodes
 - D) Different roots
 - **Answer: B**
4. A bootstrap value of 95% indicates:
 - A) Low confidence
 - B) High confidence
 - C) No confidence
 - D) Random result
 - **Answer: B**

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5. The root of a phylogenetic tree represents:

- A) Youngest organism
- B) Oldest common ancestor
- C) Protein sequence
- D) DNA molecule
- **Answer: B**

6. Sister taxa share:

- A) The same genome
- B) The most recent common ancestor
- C) Different ancestors
- D) No relationship
- **Answer: B**

7. A clade includes:

- A) Only one organism
- B) An ancestor and all descendants
- C) Different kingdoms only
- D) Random species
- **Answer: B**

8. Longer branch lengths generally indicate:

- A) Greater evolutionary change
- B) Higher protein expression
- C) More chromosomes
- D) Larger genomes
- **Answer: A**

9. Taxa refers to:

- A) Proteins

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- B) Genes only
- C) Species or sequences
- D) Amino acids
- **Answer: C**

10. Evolutionary trees are mainly used in:

- A) Comparative genomics
- B) Evolutionary biology
- C) Bioinformatics
- D) All of these
- **Answer: D**

11. Which feature measures confidence in tree branching?

- A) GMQE
- B) Bootstrap value
- C) GRAVY
- D) pI
- **Answer: B**

Practical No. 08

1. PDB stands for:

- A) Protein Data Bank
- B) Protein Database Base
- C) Public Data Bank
- D) Protein Domain Bank
- **Answer: A**

2. PDB stores:

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- A) DNA sequences only
- B) RNA sequences only
- C) Three-dimensional structures of biomolecules
- D) Protein names only

• **Answer: C**

3. PDB is:

- A) A paid database
- B) A free online database
- C) Desktop software
- D) A sequence alignment tool

• **Answer: B**

4. PDB contains structures of:

- A) Proteins
- B) Nucleic acids
- C) Biological macromolecules
- D) All of these

• **Answer: D**

5. Which AI-predicted models can also be included in PDB?

- A) AlphaFold
- B) BLAST
- C) ClustalW
- D) ProtParam

• **Answer: A**

6. PDB is mainly used for:

- A) Protein visualization
- B) Structural analysis

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- C) Molecular research
- D) All of these

• **Answer:** D

7. Simple Search in PDB can be performed using:

- A) Protein name
- B) Gene name
- C) Sequence or ID
- D) All of these

• **Answer:** D

8. By default, PDB Simple Search displays:

- A) Experimental structures
- B) Random structures
- C) DNA only
- D) RNA only

• **Answer:** A

9. Computed Structure Models (CSMs) can be:

- A) Included during search
- B) Deleted automatically
- C) Downloaded only
- D) Edited directly

• **Answer:** A

10. PDB is useful in:

- A) Drug discovery
- B) Structural biology
- C) Bioinformatics
- D) All of these

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- **Answer: D**

11. PDB mainly provides information about:

- A) Protein primary structure
 - B) Protein secondary structure only
 - C) Three-dimensional molecular structure
 - D) DNA replication
 - **Answer: C**
-

Practical No. 09

1. ProtParam is used to analyze:

- A) DNA sequences
- B) RNA sequences
- C) Protein sequences
- D) Lipid molecules
- **Answer: C**

2. ProtParam calculates protein properties using:

- A) DNA sequence
- B) Amino acid sequence
- C) RNA sequence
- D) Gene ID
- **Answer: B**

3. Which parameter indicates the total mass of a protein?

- A) GRAVY
- B) Molecular Weight
- C) Half-life

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- D) Instability Index

- **Answer: B**

4. The theoretical pI represents:

- A) Molecular size
- B) Protein length
- C) pH at which the protein has no net charge
- D) Protein stability

- **Answer: C**

5. Which parameter predicts whether a protein is stable in a test tube?

- A) Molecular Weight
- B) Instability Index
- C) Atomic Composition
- D) GRAVY

- **Answer: B**

6. An Instability Index value less than 40 indicates the protein is:

- A) Unstable
- B) Stable
- C) Hydrophobic
- D) Acidic

- **Answer: B**

7. Which parameter estimates how much light a protein absorbs?

- A) GRAVY
- B) Extinction Coefficient
- C) Half-life
- D) pI

- **Answer: B**

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8. A higher Aliphatic Index indicates:

- A) Lower stability
- B) Greater heat stability
- C) More acidic protein
- D) Lower molecular weight
- **Answer: B**

9. A positive GRAVY value indicates the protein is:

- A) Hydrophilic
- B) Hydrophobic
- C) Neutral
- D) Unstable
- **Answer: B**

10. Estimated Half-life refers to:

- A) Protein size
- B) Time required for half of the protein to degrade
- C) Protein charge
- D) Protein folding
- **Answer: B**

11. Atomic composition in ProtParam includes:

- A) Carbon, Hydrogen, Oxygen, Nitrogen, Sulfur
- B) Sodium and Potassium
- C) Calcium and Magnesium
- D) Iron and Zinc
- **Answer: A**

Practical No. 10

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1. PSIPRED is mainly used to predict:
 - A) DNA sequence
 - B) Protein secondary structure
 - C) RNA structure
 - D) Gene expression
 - **Answer: B**
2. PSIPRED requires:
 - A) DNA sequence
 - B) RNA sequence
 - C) Amino acid sequence
 - D) Chromosome number
 - **Answer: C**
3. PSIPRED predicts:
 - A) Alpha helices
 - B) Beta strands
 - C) Coil regions
 - D) All of these
 - **Answer: D**
4. The approximate Q3 accuracy of PSIPRED is:
 - A) 50%
 - B) 60%
 - C) 76.5%
 - D) 95%
 - **Answer: C**
5. MEMSAT2 is used to predict:
 - A) Protein molecular weight

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- B) Membrane protein topology
- C) Phylogenetic trees
- D) Protein families

• **Answer: B**

6. GenTHREADER is mainly used to:

- A) Find fold similarities
- B) Sequence DNA
- C) Predict molecular weight
- D) Calculate pI

• **Answer: A**

7. PSIPRED provides results in:

- A) Text only
- B) Graphical form only
- C) Both text and graphical form
- D) Audio format

• **Answer: C**

8. Which of the following is NOT predicted by PSIPRED?

- A) Alpha helix
- B) Beta strand
- C) Coil
- D) DNA replication origin

• **Answer: D**

9. PSIPRED is:

- A) A paid software
- B) A free online tool
- C) A desktop operating system

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- D) A sequencing machine

- **Answer:** B

10. GenTHREADER is useful for identifying:

- A) Distant evolutionary relationships
- B) Blood groups
- C) Restriction enzymes
- D) Chromosome numbers

- **Answer:** A

11. MEMSAT2 specializes in:

- A) Soluble proteins only
- B) Membrane proteins
- C) DNA molecules
- D) RNA molecules

- **Answer:** B

Practical No. 11

1. SWISS-MODEL is used to predict:

- A) DNA sequences
- B) RNA structures
- C) Three-dimensional protein structures
- D) Phylogenetic trees

- **Answer:** C

2. SWISS-MODEL is based on:

- A) Molecular docking
- B) Homology (template-based) modeling

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- C) PCR
- D) Gel electrophoresis

• **Answer: B**

3. SWISS-MODEL accepts protein sequences in:

- A) PDF format
- B) FASTA format
- C) DOC format
- D) JPEG format

• **Answer: B**

4. Which tools are used by SWISS-MODEL to find templates?

- A) BLAST and HHblits
- B) BLASTN and BLASTP
- C) ClustalW and MUSCLE
- D) ProtParam and PSIPRED

• **Answer: A**

5. GMQE stands for:

- A) General Model Quality Evaluation
- B) Global Model Quality Estimation
- C) Genetic Model Quality Estimate
- D) Global Molecular Quality Evaluation

• **Answer: B**

6. The GMQE score ranges from:

- A) 1–10
- B) 0–100
- C) 0–1
- D) –1 to +1

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- **Answer: C**

7. A higher GMQE score indicates:

- A) Lower model quality
- B) Better model reliability
- C) Larger protein
- D) Lower sequence identity

- **Answer: B**

8. Good homology models are usually obtained when sequence identity is:

- A) Less than 10%
- B) Greater than 35%
- C) Exactly 20%
- D) Less than 5%

- **Answer: B**

9. Which is NOT a step in SWISS-MODEL?

- A) Template Selection
- B) Sequence Alignment
- C) Model Building
- D) DNA Replication

- **Answer: D**

10. The final step in SWISS-MODEL is:

- A) Template Selection
- B) Sequence Alignment
- C) Model Evaluation
- D) DNA Sequencing

- **Answer: C**

11. SWISS-MODEL was the:

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- A) First automated homology modeling tool
 - B) First genome browser
 - C) First sequence alignment tool
 - D) First protein database
 - **Answer: A**
-

Practical No. 12

1. I-TASSER is mainly used for:
 - A) DNA sequencing
 - B) Protein 3D structure prediction
 - C) RNA extraction
 - D) PCR amplification
 - **Answer: B**
2. I-TASSER stands for:
 - A) Iterative Threading ASSEmbly Refinement
 - B) Integrated Threading Assembly Research
 - C) Iterative Template Assembly Refinement
 - D) Intelligent Threading Structure Refinement
 - **Answer: A**
3. I-TASSER predicts protein structure using:
 - A) Threading and structure refinement
 - B) Gel electrophoresis
 - C) PCR
 - D) DNA sequencing
 - **Answer: A**

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4. Which confidence score measures the reliability of the predicted model?

- A) GMQE
- B) C-score
- C) GRAVY
- D) pI
- **Answer: B**

5. The TM-score is used to measure:

- A) Protein molecular weight
- B) Structural similarity to known proteins
- C) Protein stability
- D) Protein solubility
- **Answer: B**

6. A TM-score greater than 0.5 indicates:

- A) Random similarity
- B) Correct protein structure
- C) Poor alignment
- D) Low confidence
- **Answer: B**

7. Which of the following is NOT a modeling step in I-TASSER?

- A) Threading
- B) Structural Assembly
- C) Model Refinement
- D) Restriction Digestion
- **Answer: D**

8. I-TASSER can predict:

- A) Ligand-binding sites

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- B) EC numbers
- C) GO terms
- D) All of these

- **Answer: D**

9. The B-factor in I-TASSER indicates:

- A) Molecular weight
- B) Protein flexibility
- C) Protein charge
- D) Sequence identity

- **Answer: B**

10. A TM-score less than 0.17 suggests:

- A) Correct structure
- B) High confidence
- C) Random similarity
- D) Excellent model

- **Answer: C**

11. The input required by I-TASSER is:

- A) DNA sequence
- B) Amino acid sequence
- C) Chromosome map
- D) Protein image

- **Answer: B**

Practical No. 13

1. UCSF Chimera is mainly used for:

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- A) DNA extraction
- B) 3D visualization of biomolecules
- C) PCR analysis
- D) Protein purification

- **Answer: B**

2. UCSF Chimera is:

- A) A free visualization software
- B) A paid web browser
- C) A DNA sequencing machine
- D) A protein database

- **Answer: A**

3. Chimera mainly works with which file format?

- A) PDF
- B) FASTA
- C) PDB
- D) DOCX

- **Answer: C**

4. Chimera can analyze:

- A) Hydrogen bonds
- B) Atom identification
- C) Distance and angle measurements
- D) All of these

- **Answer: D**

5. UCSF Chimera can generate:

- A) High-resolution images
- B) Animations

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- C) Movies
- D) All of these

• **Answer: D**

6. Chimera supports:

- A) Sequence–structure alignment
- B) Protein editing
- C) Interactive visualization
- D) All of these

• **Answer: D**

7. Which of the following can Chimera calculate?

- A) Partial charges
- B) Molecular distances
- C) Volumes
- D) All of these

• **Answer: D**

8. UCSF Chimera is written in:

- A) Java
- B) C++
- C) Python
- D) R

• **Answer: C**

9. Chimera is commonly used in:

- A) Structural biology
- B) Bioinformatics
- C) Molecular visualization
- D) All of these

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- **Answer: D**

10. Which of the following is NOT a feature of Chimera?

- A) Hydrogen bond analysis
- B) Sequence alignment
- C) DNA replication
- D) Image generation

- **Answer: C**

11. The primary output of UCSF Chimera is:

- A) DNA sequence
- B) Protein molecular weight
- C) Three-dimensional molecular visualization
- D) Phylogenetic tree

- **Answer: C**

Practical No. 14

1. InterPro is mainly used for:

- A) DNA sequencing
- B) Protein function analysis
- C) PCR primer design
- D) RNA extraction

- **Answer: B**

2. InterPro classifies proteins into:

- A) Families
- B) Domains
- C) Functional sites

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- D) All of these

• **Answer: D**

3. InterPro combines predictive models known as:

- A) Templates
- B) Signatures
- C) Fragments
- D) Reads

• **Answer: B**

4. Which of the following is a member database of InterPro?

- A) Pfam
- B) PROSITE
- C) SMART
- D) All of these

• **Answer: D**

5. A protein domain is:

- A) A functional or structural unit
- B) A chromosome
- C) A DNA fragment
- D) An RNA molecule

• **Answer: A**

6. A protein family consists of:

- A) Unrelated proteins
- B) Related proteins with a common origin
- C) DNA sequences only
- D) Different organisms only

• **Answer: B**

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7. Homologous superfamilies contain proteins with:

- A) Identical sequences only
- B) Similar structures but low sequence similarity
- C) No structural similarity
- D) Different amino acids only
- **Answer: B**

8. Repeats in InterPro are:

- A) Long chromosomes
- B) Short recurring sequence patterns
- C) DNA mutations
- D) Protein databases
- **Answer: B**

9. Which is an example of a functional site in InterPro?

- A) Active site
- B) Binding site
- C) PTM site
- D) All of these
- **Answer: D**

10. Unintegrated entries are:

- A) Deleted entries
- B) Entries not yet fully curated or integrated
- C) Incorrect protein sequences
- D) DNA sequences only
- **Answer: B**

11. InterPro helps predict:

- A) Protein domains

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- B) Protein families
- C) Functional annotation
- D) All of these
- **Answer: D**

MOST REPEATED MCQS IN PAST PAPERS:

Q1. In what formats can data be exported from GenBank?

- A. TXT (.txt) file
 - B. FASTA file
 - C. XML file
 - D. **All of the above**
-

Q2. GenBank and Swiss-Prot are examples of:

- A. **Primary databases**
 - B. Secondary databases
 - C. Composite databases
 - D. None of the above
-

Q3. Which of the following statements about Global Alignment is incorrect?

- A. It aligns two sequences from end to end.
 - B. It is suitable for comparing sequences of similar length.
 - C. It is performed using the Needleman–Wunsch algorithm.
 - D. **It identifies only the most similar local regions between two sequences.**
-

Q4. Which of the following is NOT a step of the ClustalW algorithm?

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- A. Pairwise sequence alignment
 - B. Construction of a distance matrix
 - C. Guide tree construction using the Neighbor-Joining method
 - D. **Prediction of protein secondary structure**
-

Q5. Suppose your lab partner is using the BLAST program and the best E-value obtained is 3. What does this indicate?

- A. The match is highly significant.
 - B. **Approximately three matches with a similar score are expected to occur by chance.**
 - C. The sequence is an exact match.
 - D. The alignment has no similarity.
-

Q6. The time required to generate a phylogenetic tree depends on:

- A. Number of sequences
 - B. Length of sequences
 - C. Tree-building algorithm used
 - D. **All of the above**
-

Q7. Amino acid composition denotes the _____ of individual residues in a protein.

- A. **Total number**
 - B. Strength
 - C. Concentration
 - D. Charge
-

Q8. GMQE (Global Model Quality Estimation) is expressed as a value between:

- A. 0.5 to 1
- B. **0 and 1**
- C. Less than 0
- D. Greater than 1

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Q9. UCSF Chimera can be used to delete protein chains and non-standard residues.

- A. **True**
- B. False

Q10. In what format can data be exported from GenBank?

- A. TXT (.txt) format
- B. FASTA format
- C. XML format
- D. **All of the above**

Q11. Which of the following tools is provided on the UniProt website for protein sequence analysis?

- A. BLAST
- B. Align
- C. Retrieve/ID Mapping
- D. **All of the above**

Q12. Which algorithm is commonly used for the alignment of DNA sequences?

- A. K-Means Algorithm
- B. **Needleman–Wunsch Algorithm**
- C. Dijkstra Algorithm
- D. Huffman Algorithm

Q13. Which of the following is NOT a step of the ClustalW algorithm?

- A. Pairwise sequence alignment
- B. Construction of a distance matrix

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- C. Guide tree construction using the Neighbor-Joining method
D. **Prediction of protein secondary structure**
-

Q14. _____ are sequences used for recognizing an alignment in the MEGA software.

- A. Conserved sequences
B. Query sequences
C. **Aligned sequences**
D. Random sequences
-

Q15. MEGA software allows users to save or export a generated phylogenetic tree in which format?

- A. Newick (.nwk)
B. PDF/Image format
C. MEGA session format (.mtsx/.meg depending on version)
D. **All of the above**
-

Q16. ProtParam computes different parameters using:

- A. DNA sequence
B. RNA sequence
C. **Protein (amino acid) sequence**
D. Chromosome sequence
-

Q17. Which of the following software is used to visualize and analyze the three-dimensional (3D) structure of proteins?

- A. BLAST
B. SWISS-MODEL
C. InterPro
D. **UCSF Chimera**

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Q18. What is the full form of MEGA?

- A. Molecular Evolutionary Genetic Analysis
- B. Molecular Evolution and Genome Analysis
- C. **Molecular Evolutionary Genetics Analysis**
- D. Molecular Evolutionary Gene Alignment

Good Luck 😊

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